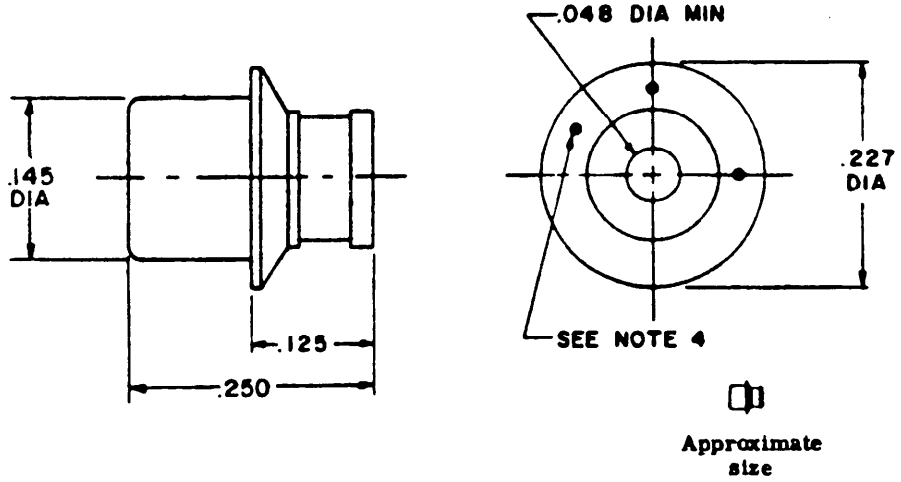
MIL-C-11015/29C(USAF)
30 July 1974
SUPERSEDING
MIL-C-11015/29B(USAF)
6 June 1974

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), (FEED-THRU OR STAND-OFF) STYLES CK74, CK75, CK78, CK79 AND CK85

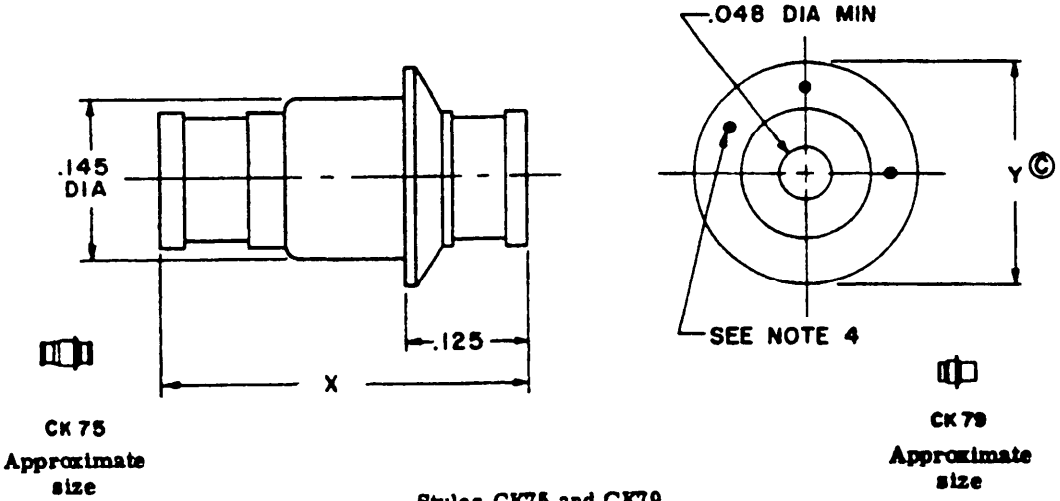
This specification is approved for use by Aeronautical Systems Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.



Approximate size

Style CK85



CK 75
Approximate size

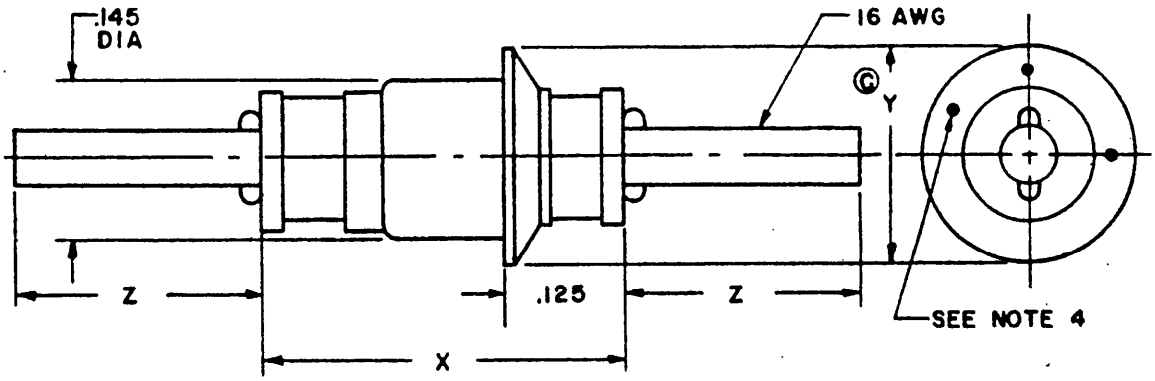
CK 79
Approximate size

Styles CK75 and CK79

© Denotes changes.

FSC 5910

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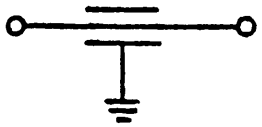


CK 74
Approximate
size



CK78
Approximate
size

Styles CK74 and CK78



CIRCUIT DIAGRAM

STYLE	X	© Y	Z
CK74	.380	.227	.250
CK75	.380	.227	---
CK78	.250	.195	.125
CK79	.250	.195	---

INCHES	MM	INCHES	MM
.048	1.22	.227	5.77
.125	3.18	.250	6.35
.145	3.68	.380	9.65
.195	4.95		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.015 (.38 mm).
4. Color dots indicating capacitance value read clockwise starting with 10 o'clock dot. Dots to be located at approximately 10, 12, and 3 o'clock.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See above figure.
Case type - Feed-thru or standoff (solder mount).
Capacitance value - See table I.
Capacitance tolerance - ±20% (M).
Rated temperature - -55°C to +125°C.
© Rated voltage - Styles CK74 and CK75, 500 volts dc, to +85°C, styles CK78, CK79, and CK85, 300 volts dc, to +85°C. Linearly derate to 50 percent at +125°C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 750 ±20 volts dc applied between the terminal and the mounting flange.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition B (50,000 ft).
Test potential - 750 ±20 volts dc applied between the terminal and the mounting flange.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 10,000 megohms, min, measured between the terminal and the mounting flange.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency - 1.0 ±0.1 kHz.
Test voltage - 1.0 ±0.2 volt rms.

Dissipation factor (DF): 3.0%, max.

Shock (specified pulse): Method 213 of MIL-STD-202, condition A (50 G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15 G).
Mounting - Capacitors shall be rigidly mounted by the body.

Thermal shock and immersion:
Thermal shock: Method 107 of MIL-STD-202, condition A.
Immersion: Method 104 of MIL-STD-202, condition B.
IR - 1,000 megohms, min.

Salt spray (corrosion): Not applicable.

Terminal strength: Not applicable.

Moisture resistance: Method 106 of MIL-STD-202.
IR - 1,000 megohms, min.
ΔCap. - Within ±10% of initial measurement.

Solderability: Method 208 of MIL-STD-202.
1 terminal.

Resistance to soldering heat: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-11015 (symbol BX).

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202.
2,000 hours (for qualification); 250 hours (performance check); 1,750 hours continuation test).
Test potential - 200% of rated voltage.
IR - 2,500 megohms, min (at 85°C);
5,000 megohms, min (at 25°C).
DF - 5.0% max (at 25°C).

Marking: Capacitors shall be marked as shown on the figure (marking shall conform to method II of MIL-STD-1285). The type designation, date and source codes, manufacturer's name or trademark, capacitance, and dc rated voltage shall be marked on the package.

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© TABLE I. Styles CK74, CK75, CK78, CK79, and CK85 characteristics.

Type designation	Capacitance value
	(pF)
CK--BX100M	10
CK--BX150M	15
CK--BX220M	22
CK--BX330M	33
CK--BX470M	47
CK--BX680M	68
CK--BX101M	100
CK--BX151M	150
CK--BX221M	220
CK--BX331M	330
CK--BX471M	470
CK--BX501M	500
CK74BX681M	680
CK75BX681M	680
CK78AW681M	680
CK79AW681M	680
CK85AW681M	680
CK74BX102M	1,000
CK75BX102M	1,000
CK78AW102M	1,000
CK79AW102M	1,000
CK85AW102M	1,000
CK7-AW152M	1,500
CK7-AW182M	1,800
CK74AW222M	2,200
CK75AW222M	2,200
CK74AW332M	3,300
CK75AW332M	3,300

Preparing activity:
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(Project 5910-F258)

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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